

Padmanabhan Vijendran

Big Data Developer / Data Engineer

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Professional Profile

- **Senior Backend & Distributed Systems Engineer** with **14 years of experience** specializing in designing and developing highly scalable, low-latency applications within the financial and enterprise sectors.
- **Fintech Domain Expertise:** Proven experience in orchestrating complex, event-driven systems (Payments and Regulatory Reporting), leveraging **Java** and modern big data technologies.
- **Technical Mastery:** Extensive expertise in the **JVM ecosystem** (Java 8/17, Spring), **Apache Spark** (including **Structured Streaming**), **Kafka**, and NoSQL/Relational databases (**HBase**, PostgreSQL).
- **Cloud & DevOps:** Proficient in designing cloud-native solutions, CI/CD automation (**Jenkins**, **GitHub Actions**), containerization (**Docker**, **Kubernetes**), and holding a **Google Cloud Certified - Professional Data Engineer** certification.
- **Adaptable & Analytical:** Strong analytical and problem-solving skills, with a focus on writing clean, well-tested code, demonstrating continuous growth (**Python**, **Machine Learning**, **AI Tools**).

Professional Experience

Lead Data Engineer, *Royal Bank of Canada*

06/2023 – Present

Project Description: *EODc Regulatory and Compliance Automation*

Toronto, Canada

This initiative focuses on designing and developing a comprehensive framework to automate the generation of critical regulatory and End-of-Day Contract (EODc) reports for RBC's US subsidiaries, including Georgia Bank and City National Bank. As part of this effort, I also spearheaded the creation of the EODc Compliance Checker, a key internal tool that validates daily contract reports to ensure they meet both industry and internal standards. The project leverages a modern technology stack to build robust, scalable, and secure applications that support key business operations and regulatory compliance.

Roles and Responsibilities:

- **Engineered** a scalable, end-to-end data solution using **Java** and **Apache Spark** to automate the generation of critical regulatory and EODc reports.
- **Architected and developed** the **EODc Compliance Checker**, a Java-based application that validates complex daily contract reports and **ensured compliance with internal and industry standards**.
- **Modernized** application components by migrating from **Java 8 to Java 17** and **Spring 4 to Spring 6**, improving performance and ensuring compatibility with modern standards.
- **Led** the migration of **CI/CD pipelines from Jenkins to GitHub Actions**, **streamlining deployment workflows** and **reducing deployment time**.
- **Managed** and **resolved** high-priority **P1/P2 security vulnerabilities** across multiple applications, maintaining compliance with organizational and industry standards.
- **Optimized** data processing for voluminous **EODc data packages, exceeding 200GB**, using multi-threading techniques to ensure high performance.

- **Maintained** a high level of code quality by implementing **unit tests with over 90% coverage**, integrating **Sonar Lint** for static analysis, and developing **functional test cases** with the **Cucumber Framework**.
- **Collaborated** with **Agile** teams and **business analysts** to translate complex business requirements into technical solutions and ensure the timely delivery of reports.

Senior BigData Developer, *Cognizant Technology Solutions - Client: Ford*

08/2021 – 06/2023
Mississauga, Canada

Project Details: Ford engineers use reports generated by an **ETL** framework to design future cars with low manufacturing costs. We are migrating the **ETL** framework from **Sqoop** and **Hive** to **Apache Spark Structured Streaming** so that engineers can access near **real-time** reports.

Roles and Responsibilities:

- Migrated **ETL** framework from **Sqoop/Hive** to **Apache Spark Structured Streaming** for Ford Engineering.
- Developed **Spark Structured Streaming** application using **pyspark** that **reduced data processing time** and increased **report accuracy**.
- Implemented **ETL** pipelines using **Apache Spark, Kafka** and **MongoDB** that enabled engineers to utilize near real-time reports for designing future cars at low manufacturing cost.
- Designed and developed **Java** application to efficiently fetch data from LDAP servers.
- Refactored **Java** applications to improve readability and performance
- Explored new technologies using **POCs** and documented the approaches to implement new project requirements

Senior BigData - Java Developer, *Kelly Services - Client: Credit Agricole CIB*

03/2018 – 07/2021
Singapore

Project Details: Cash Management Transactions (CMT) - Real-Time Payment Orchestration

Engineered and implemented a **high-performance, low-latency Internal Payment Engine (IPM)** designed to orchestrate the entire lifecycle of incoming and outgoing **SCT** and **Cross-Border** transactions. The resilient, **event-driven architecture**, built using **Java 8, Apache Spark Structured Streaming, Kafka**, and **HBase**, managed communication with external banking hubs and numerous internal systems (e.g., Ledger, Billing), ensuring real-time integrity and finality of payments. Delivered using **Agile methodologies**.

Roles and Responsibilities:

- **Engineered a scalable, real-time internal payment engine (IPM)** using **Java 8** and **Apache Spark Structured Streaming** to orchestrate high-volume payment transactions and directly support critical business operations.
- **Designed and built a resilient, event-driven data pipeline** with **Kafka** and **HBase** that enabled the real-time ingestion, storage, and analysis of transaction data, ensuring data integrity for downstream systems.
- **Solved a critical integration challenge** by developing **custom source and sink connectors for IBM MQ** using Spark Structured Streaming, which streamlined data ingestion and eliminated the need for an external middleware component.
- **Developed a versatile data translation layer** that enabled seamless communication between the payment engine and various internal applications by **translating proprietary formats** (fixed-width, JSON) into a standardized messaging structure (ISO 20022 XML).

- **Enhanced system observability and quality** by setting up application monitoring dashboards using the **ELK stack** and automating functional tests with the **Cucumber** framework, providing real-time visibility into performance and ensuring high code quality.
- **Streamlined the CI/CD pipeline** by developing **Jenkins pipelines** with Groovy and shell scripting, automating the build, test, and deployment of code changes.
- **Automated development tasks** by creating **Python scripts** to generate performance reports and demo tools, improving overall team efficiency.

Big Data Developer, Cognizant Technology Solutions - Client: Ford

02/2016 – 11/2017

Project Details: I worked on the project that generated reports to help Ford Engineer to design a future cars at low manufacturing cost. We implemented **ETL** using **BigData** tools such as **Apache Spark 2.3**, **Hive**, **Sqoop** and **Oozie**.

Chennai, India

Roles and Responsibilities:

- Extracted data from various Oracle databases to **HDFS** and **Hive** using **Sqoop** scripts
- Transformed data to the required schema using **Hive** scripts
- Developed **Java** application to generate Excel reports from **Hive** table data using **Apache POI**
- Explored ETL with **Alteryx**, **Apache Spark** using **POCs**
- Automated manual tasks using **Python** and **shell scripts**

IT Analyst, Tata Consultancy Services - Client: Target, USA

01/2011 – 01/2016

Project Details: I worked on the project that processed product images and details for Target website using **ETL Java** application and **SQL** scripts.

Chennai, India

Roles and Responsibilities:

- Developed **ETL Java** application to process product information using **Map-Reduce**
- Wrote **SQL** scripts to transform product related data
- Automated manual tasks using **Python**
- Investigated and analyzed application **support incidents** through to root cause and fix
- Proactively monitored services and responded to alerts raised by customers and **monitoring** systems, ensuring the availability and performance of the **Teamsite** Content Management Systems

Technical Skills

Core Languages

Java 17, Python 3, Shell Scripting

Backend Frameworks

Spring Framework 6 (Core, DI, Spring Data JPA, Spring Security)

DevOps & CI/CD

Docker, Jenkins, GitHub Actions, Artifactory, UrbanCode Deploy

Cloud Platforms

AWS, Azure, GCP, Oracle Cloud (Certified Google Cloud Professional Data Engineer)

Distributed Systems

Apache Spark (Structured Streaming), Kafka, Hadoop, Hive

Databases & Storage

HBase, PostgreSQL, MySQL, MongoDB

Testing & Quality

JUnit-5, Mockito, Cucumber

AI/ML

Python Libraries (pandas, scikit-learn), Algorithms (KNN, Regressions, Random Forest, XGBoost, SVM), AI Tools (Github Copilot, Claude Code, Gemini CLI, LangChain)

Certificates

- Google Cloud Certified - Professional Data Engineer [↗](#)
- Microsoft Certified: Azure AI Fundamentals [↗](#)
- AWS Cloud Practitioner [↗](#)

Personal Projects

AI Chat Exporter Chrome Extension [↗](#)

Developed a browser extension to export full chat conversations from Gemini and ChatGPT into a clean, portable Markdown file, developed through self-guided learning and rapid prototyping.

Technologies: JavaScript, HTML, CSS, Chrome Extension API.

Interactive ChatGPT Command Line Interface (CLI) [↗](#)

Created a CLI tool for an interactive conversational experience with the OpenAI API, featuring real-time response streaming.

Technologies: Python 3, OpenAI API, command-line interface design.

Machine Learning Odyssey [↗](#)

A portfolio of hands-on Machine Learning projects, including a breast cancer classification model and a house sales price prediction model, showcasing proficiency in data analysis and model development.

Technologies: Python, Scikit-learn, Pandas, NumPy, XGBoost, Jupyter Notebook.

Courses

GCP Data Engineering, *Google Qwik-Labs* [↗](#)

Education

Bachelor of Engineering - Electronics and Communication, *Anna University*

2006 – 2010
Chennai, India